

Parallel Decoder Transformer: Planner-Conditioned Latent Coordination for Model-Intrinsic Parallel Generation

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Abstract

Autoregressive language models expose one causal token frontier, even when the requested document contains sections that could be developed concurrently. Existing parallel-generation systems arrange external branches around an otherwise unchanged model. We instead formulate *model-intrinsic parallel generation*: a single trained architecture owns multiple causal frontiers and produces one next-token distribution for each frontier in every synchronized decoding round. The Parallel Decoder Transformer (PDT) retains a frozen shared lower knowledge trunk and replaces the upper trunk with three independently parameterized physical decoder stacks. A prompt-time set planner produces three unordered continuous outlines, each hard-routed to one decoder as persistent Plan-KV memory, while a finite product-quantized notes bus carries block-delayed latent messages among the decoders. Autoregression is preserved within each lane; same-round lane tokens are conditionally independent given the source, plans, private histories, and previously committed messages. We specify source-grounded supervision for long-form historical exposition with single-owner cited facts and token-aligned cross-lane dependencies, a composite objective, a staged curriculum, and preregistered causal evaluations: plan swap and removal, delayed-message ablation, a parameter-matched self-only control, dependency-token likelihood, and blinded human fact audits. The architecture and evaluation pipeline are implemented; scientific training and held-out evaluation are in progress. This paper presents the theory, design, and falsifiable protocol, not a positive empirical result.

1 Introduction

A decoder-only Transformer represents one sequence distribution, $p(y | x) = \prod_t p(y_t | x, y_{<t})$, and therefore exposes one next-token decision at a time. That factorization is appropriate when the next passage depends on all preceding prose. It is unnecessarily restrictive when the response decomposes into complementary sections whose local prose is sequential but whose development is partly concurrent, as in long-form histories, operating procedures, and technical briefings.

Prior systems exploit such structure from outside the model. Skeleton-of-Thought elicits a textual skeleton and expands its points through parallel calls or batched branches [Ning et al., 2024]; APAR trains a model to emit fork control tokens and restricts attention over the resulting paragraph tree [Liu et al., 2024]; PASTA learns an annotation language with asynchronous promises and explicit synchronization [Jin et al., 2025]. These systems show that generation need not follow the final reading order. They leave open the architectural question studied here: can the model itself contain several persistent, physically distinct causal frontiers, learn which part each frontier should write, and exchange only a bounded latent summary of completed work?

We call this setting *model-intrinsic parallel generation*. “Intrinsic” does not mean computation is free or that concurrency guarantees a wall-clock speedup. It means that planning, semantic assignment, frontier state, and cross-frontier communication are learned components of one model rather than a controller issuing independent language-model requests: the model returns three next-token distributions in one grouped

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forward operation, and each frontier owns private upper-layer parameters and KV state, so decoder identity is a physical axis rather than a natural-language role label.

The design makes four contributions.

1. A causal factorization for three synchronized, lane-local autoregressive sequences: same-round tokens never observe one another, and cross-lane influence flows only through messages committed at an earlier block boundary.
2. An architecture realizing that factorization: a frozen shared lower knowledge trunk, a continuous unordered set planner, and three independently parameterized upper decoder stacks whose private caches and persistent hard-routed Plan-KV make assignment structural and inspectable.
3. A finite delayed communication channel: every completed 32-token lane block may publish four 8-bit product-VQ indices. Producer and lag headers address the message; the learned code tuple is the entire payload.
4. Source-grounded data, losses, curriculum, controls, and causal gates that separate four questions: can the executor follow oracle plans, does the planner learn decomposition, does plan assignment cause fact ownership, and do sibling messages help exactly where delayed dependencies require them.

2 Problem Formulation

2.1 Three sections, not three guesses

Let x contain a source document and an expository request. The required output is a tuple of three multi-paragraph sections $Y = (y^1, y^2, y^3)$ and a presentation permutation $\pi \in S_3$; the reader sees $y^{\pi_1} \| y^{\pi_2} \| y^{\pi_3}$. There is no fourth synthesis decoder and no serial rewriting pass: the sections are complementary parts of one answer, not independent samples from which a best response is selected.

The planner maps x to three semantic plans $Z = \{z^1, z^2, z^3\}$, where each $z^k = (z_1^k, \dots, z_N^k)$ is an ordered outline of at most N nodes. The lane-level collection is an unordered set: a physical decoder has no permanent role such as “background” or “consequences,” and presentation order is predicted separately from physical address. Each source fact f has exactly one owner lane $o(f)$; another lane may be licensed to reference it, but a fact written only by the wrong lane does not satisfy ownership. This rules out the weak objective in which the union of three outputs receives credit regardless of who wrote what.

2.2 Block-delayed causal factorization

Generation is divided into blocks of τ lane-local tokens (canonically $\tau = 32$). Let $y_{b,r}^k$ be token r of block b in lane k , let $y_{<b,r}^k$ be all earlier tokens in that lane, and let $\mathcal{M}_{<b}$ be the messages whose publication time plus delivery delay is no later than block b ; with a one-block delay, a message written after block $b - 1$ first affects block b . The model uses the factorization

$$p_{\Theta}(Y, Z, \pi \mid x) = p_{\phi}(Z, \pi \mid x) \prod_b \prod_{k=1}^3 \prod_{r=1}^{|y_b^k|} p_{\Theta,k}(y_{b,r}^k \mid x, z^k, y_{<b,r}^k, \mathcal{M}_{<b}). \quad (1)$$

For a fixed synchronized microstep (b, r) , the three current tokens are conditionally independent given the right side of Equation 1. The sections are not globally independent—their dependence is mediated by the common source, the jointly produced plan set, and previously committed messages—but same-round communication is prohibited, because it would impose an order among lanes and destroy the claimed concurrency.

3 Parallel Decoder Transformer

3.1 Overview and frozen shared trunk

Figure 1 shows the complete topology; Figure 2 expands one physical upper layer. A revision-pinned decoder-only Transformer is split at layer L_f . The embedding and layers below L_f , $T_{\theta_{\text{io}}}$, form a single frozen trunk:

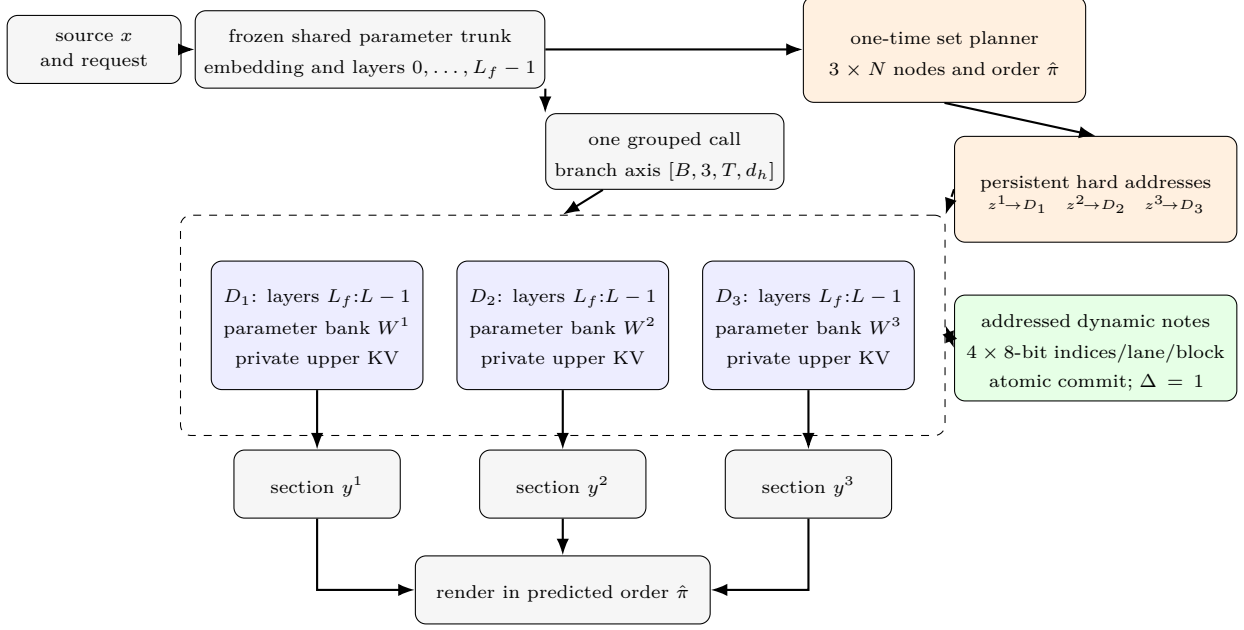


Figure 1: System topology. Solid arrows carry token states or planner computation; dashed arrows carry persistent plan memory or block-delayed latent messages. The three upper stacks advance in one grouped operation and share neither base weights nor upper-layer KV caches.

the common prompt is prefilled once below the fork and the cache is expanded onto three lane rows. Layers at and above L_f are copied into three independently trainable physical stacks, each receiving its lane’s persistent outline and the visible dynamic-note window through separate cross-attention residuals. Parameter sharing below the fork is not a shared hidden trajectory: once the lanes emit different tokens, their rows pass through the same lower weights in one grouped batch but keep separate cache rows and hidden states.

3.2 Continuous set planner

The planner runs once, before generation. Normalized prompt states $H^x = T_{\theta_{\text{io}}}(x)$ are projected from the trunk width d_h to planner width d_p , and three groups of N learned queries cross-attend to H^x through a small Transformer decoder:

$$\hat{Z}, \hat{V}, \hat{s} = P_\phi(H^x), \quad \hat{Z} \in \mathbb{R}^{3 \times N \times d_p}, \quad (2)$$

where $\hat{V}$ holds node-validity logits and $\hat{s} \in \mathbb{R}^3$ holds presentation scores. The three lane groups are trained as a set: the loss enumerates all six lane permutations and differentiates only through the minimum-cost assignment, so set-valued targets acquire no meaning from storage order [Lee et al., 2019], while node order within a lane remains meaningful. The planner never consumes generated tokens and is never re-invoked, so no hidden serial planner loop reintroduces one-step-at-a-time coordination.

3.3 Three physical decoder stacks

The pretrained upper layers $T_{\theta_{\text{hi}}}^{L_f:L-1}$ are cloned into $D_{\theta_1}, D_{\theta_2}, D_{\theta_3}$. Every self-attention projection, MLP projection, and RMS normalization weight carries a leading physical-decoder axis, $W \in \mathbb{R}^{3 \times d_{\text{out}} \times d_{\text{in}}}$, whose slices are initialized identically but are distinct parameters with independent gradients, advanced by grouped matrix multiplication. Each upper attention layer stores private keys and values with shape $[B, 3, H_{\text{kv}}, T, d_{\text{head}}]$. This differs materially from three prompt labels passed through a shared decoder: the private cache axis, branch parameter slice, and hard-routed plan memory already determine the address, so no lane needs to infer its assignment from a tag inside a shared cache.

3.4 Persistent hard-routed Plan-KV

Planner nodes are projected from d_p to a 256-wide memory representation. At every physical upper layer ℓ , lane k reads only its own valid nodes:

$$\tilde{z}_n^k = W_z z_n^k + e_n^{\text{node}}, \quad (3)$$

$$A_{\text{plan}}^\ell(h^k, z^k) = W_O^\ell \text{softmax} \left(\frac{(W_Q^\ell h^k)(W_K^\ell \tilde{z}^k)^\top}{\sqrt{d_a}} \right) W_V^\ell \tilde{z}^k. \quad (4)$$

The result enters through a learned gated residual. Nodes remain individually addressable as keys and values—mean pooling into a single additive plan vector is not the canonical path—and plan memory is read-only and persists for the complete generation.

3.5 Finite delayed dynamic notes

Static plans define intended work; dynamic notes summarize work actually completed. At the end of block b , lane k ’s block-final hidden state is projected to $u_b^k \in \mathbb{R}^{256}$, split into $M = 4$ subvectors, and quantized:

$$c_{b,m}^k = \arg \min_{c \in \{1, \dots, C\}} \|u_{b,m}^k - e_{m,c}\|_2^2, \quad M = 4, \quad C = 256. \quad (5)$$

The transmitted payload is the tuple $(c_{b,1}^k, \dots, c_{b,4}^k)$, with capacity $M \log_2 C = 32$ bits per lane block; receivers decode the indices through the learned codebooks [van den Oord et al., 2017]. Every note carries non-payload headers identifying its producer and age: Shared Notes Cross-Attention (SNC) adds learned producer, message-kind, and log-lag embeddings before forming keys and values, and the receiver’s hidden state forms the query, so the same addressed note can be used differently by each lane. The window retains up to 16 blocks. Writes are atomic at a block boundary: a note published at block b is invisible during block b and becomes readable at block $b + 1$, and all three writes commit before any receiver begins the next block. Both Plan-KV and SNC use zero-initialized output projections and closed residual gates, so the untrained extension starts as the cloned pretrained decoder rather than an uncontrolled perturbation.

3.6 Synchronized parallel generation

Natural generation follows one canonical loop:

1. encode the prompt, run the planner once to resolve three valid plan memories and one presentation order, prefill the common prompt below the fork once, and expand the lower cache to three rows;
2. at each round, execute one grouped lower-plus-upper forward with the assigned Plan-KV and sample one token from each active lane’s returned distribution;
3. when every active lane reaches a 32-token boundary, compute all product-VQ writes, commit them together, and only then build the next block’s addressed note windows;
4. a lane that emits EOS becomes logically inactive but remains a masked physical row in the packed cache; other lanes continue without falling back to separate model calls, and the finished sections are decoded independently and concatenated in predicted presentation order.

Figure 3 makes the synchronization and message delay explicit: a vertical slice through the three lane rows is one grouped decoding round, and no lane can observe a sibling’s token or block-end code from the current block.

The loop emits up to three sampled tokens per synchronization round while preserving token-level autoregression inside every section. For final lane lengths L_1, L_2, L_3 , a serial concatenated decoder requires $R_{\text{serial}} = \sum_k L_k$ decode rounds; the packed frontier has algorithmic span $R_{\text{PDT}} = \max_k L_k$ after planning and prefill, for a maximum span reduction of

$$\rho_{\text{span}} = \frac{\sum_{k=1}^3 L_k}{\max_k L_k} \leq 3. \quad (6)$$

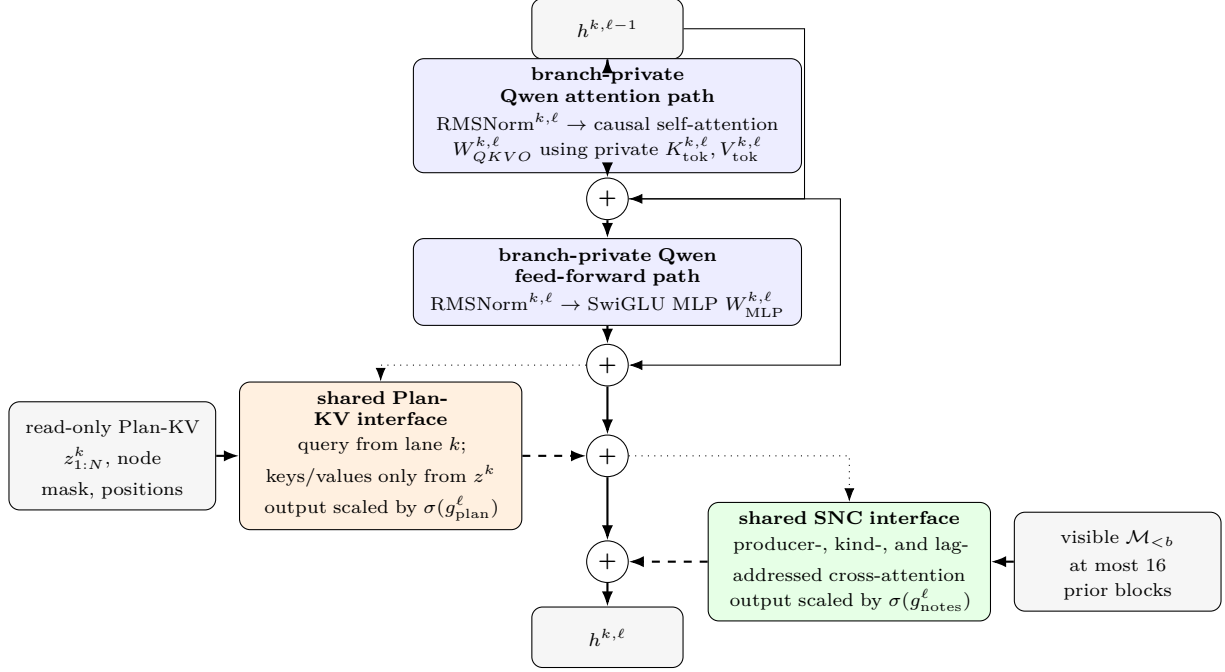


Figure 2: Anatomy of upper layer ℓ for physical lane k , repeated at every physical upper layer. The Qwen self-attention, normalization, MLP weights, and token KV cache are lane-private; Plan-KV and SNC have shared interface parameters but operate on lane-private queries and structurally addressed memories. Dotted arrows carry the lane state into the cross-attention queries; dashed arrows carry memory inputs and gated residual outputs.

Equation 6 is a dependency-depth statement, not a hardware speed claim: the upper weights are triplicated, and real latency depends on kernels, memory traffic, occupancy, cache length, and lane balance. We therefore report synchronization depth, arithmetic work, memory footprint, and measured wall time separately.

4 Source-Grounded Long-Form Supervision

4.1 Domain and source ingress

Short answers and question-answer pairs do not test the mechanism: a lane can satisfy them before persistent plans, ownership, and delayed coordination become meaningful. The first study therefore uses historical events and processes whose sources naturally support several long complementary sections; every teacher section must contain 700–1,000 Qwen tokens and four to twelve connected paragraphs, and lists, timelines, biographies as primary topics, and catalog-like pages are excluded. Historical text also supplies checkable temporal constraints. “Natural” in the data contract means that a record’s presentation order and cross-section references obey the document’s causal and expository prerequisites, not that the record proves latent coordination.

Each source begins from one exact English Wikimedia revision, retained as an immutable raw bundle (wikitext, rendered semantic HTML, assessments, reference records, revision identifiers, licenses, digests). Only a derived representation is model-visible: a strict DOM allowlist keeps the article title, hierarchical content headings, and ordinary prose paragraphs, with inline links contributing only display text; tables, lists, infoboxes, figures, citation markers, navigation, and all other non-prose apparatus are excluded, and reference identities remain audit metadata never concatenated into the student prompt. Articles are accepted or rejected whole—never truncated or stitched—under the preregistered gates of Table 1.

one grouped round r : three next-token distributions

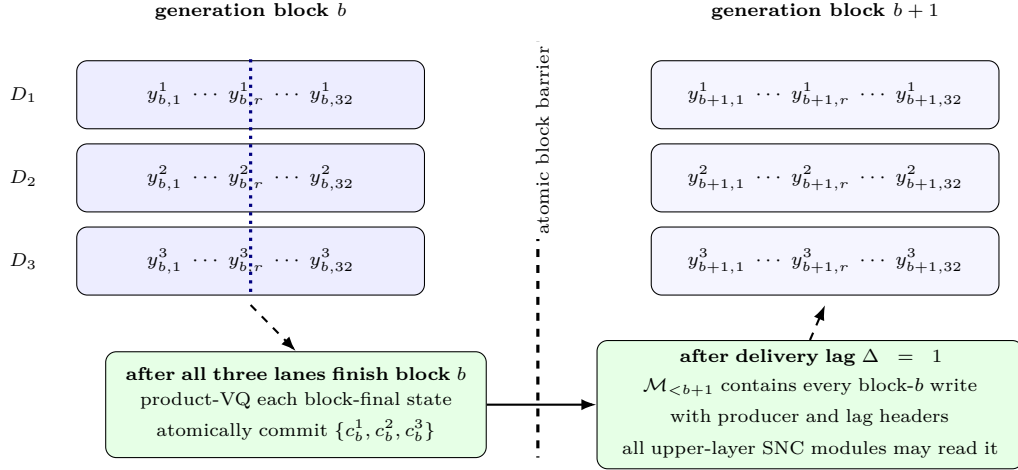


Figure 3: Synchronized causal timeline for block size $\tau = 32$. Each microstep advances all active physical lanes in one grouped call; after the final microstep, the three product-VQ writes are committed as one atomic set, cannot affect block b , and first enter the addressed SNC windows at block $b + 1$.

Gate	Requirement
Topic and age	English main-namespaced event or process; non-redirect, non-disambiguation; event ended at least 25 years before acquisition.
Assessment	FA, GA, A, or B from a relevant history-oriented WikiProject; no citation, neutrality, accuracy, original-research, cleanup, or hoax maintenance template.
Complete cleaned body	3,000–7,000 pinned-Qwen tokens, at least six substantive sections, at least twelve substantive paragraphs.
Citations	At least 30 inline reference occurrences, 15 distinct cited works, eight scholarly or institutional sources; citations on at least 70% of substantive paragraphs; no cited work above 25% of occurrences.
Split integrity	Related pages and manually reviewed family keys assigned to one train, validation, or test split before teacher generation.

Table 1: Whole-article source gates for the historical corpus.

4.2 Two-stage teacher construction

Teacher data are produced in two schema-constrained stages.

Stage A: cited facts. For each accepted source, the teacher extracts 18–48 atomic facts spanning at least twelve source paragraphs and four source sections. Every fact carries an exact source quote, character offsets, paragraph ID, and local reference IDs that must agree with the immutable source record, plus one plausible hard negative that alters an entity, date, quantity, or relation.

Stage B: joint three-lane answer. One request receives the complete source and validated fact inventory and returns one expository prompt, exactly three plans, all three target sections, and their presentation order—jointly, because cross-lane ownership and references are properties of the complete document. Every plan has four to eight nodes specifying its expository objective, owned facts, permitted reference facts, dependencies, and target paragraphs. Every source fact has exactly one OWNER lane; each other lane labels it REFERENCE or ABSENT, an OWNER or REFERENCE label requires an exact quote in the target section, and hard negatives must remain absent everywhere. Cross-lane references are optional: when present, a dependency names the owner plan, node, and quote and the receiver node and quote; the owner must

precede the receiver in presentation order; and the owner quote must fall in a generation block visible at least one block before the first receiver token. Zero-dependency records are valid—the compiler does not invent communication to make the notes bus look useful.

4.3 Tokenized training record

Retokenization uses the exact student tokenizer and appends one EOS token to each target, so per-lane stopping is learned rather than bolted onto inference. A frozen 1,024-dimensional BGE encoder [Xiao et al., 2023] embeds every true fact and hard negative and a textual form of every outline node; a fixed Gaussian Johnson–Lindenstrauss projection maps node targets to 512 dimensions. Each record materializes the fact embeddings, padded continuous plan targets with node-validity masks, exact fact-to-node routing, one active outline node per 32-token block, OWNER/REFERENCE/ABSENT labels at the exact lane and block where evidence occurs, exact dependency-token masks with source-block-to-target-token edges verified against the one-block delay, and the reader-facing presentation ranks. The first pilot targets 128 accepted records balanced across eight historical categories, each manually inspected; a 2,048-record corpus is the architecture gate. Four separately curated examples currently exercise the complete schema and trainer interface as data-contract inspection records, not a training corpus.

5 Learning Objective

5.1 Unordered plan matching

For example i , let $\hat{z}_{k,n}$ and $\hat{v}_{k,n}$ be predicted plan nodes and validity logits, and let $z_{k,n}$ and $v_{k,n}$ be teacher targets. For each lane permutation $\sigma \in S_3$, define

$$C_i(\sigma) = \frac{\sum_{k,n} v_{k,n} [1 - \cos(\hat{z}_{\sigma(k),n}, z_{k,n})]}{\sum_{k,n} v_{k,n}} + \frac{1}{3N} \sum_{k,n} \text{BCE}(\hat{v}_{\sigma(k),n}, v_{k,n}). \quad (7)$$

The minimum-cost assignment $\sigma_i^* = \arg \min_{\sigma \in S_3} C_i(\sigma)$ is selected discretely and detached; gradients flow through the selected predicted nodes, and all downstream semantic axes are expressed in matched order.

5.2 Execution and semantic losses

Lane token likelihood. $\mathcal{L}_{\text{tok}}$ is exact teacher-forced next-token cross entropy over every valid token, including each lane’s EOS; the first token of every target block is scored from the preceding prompt- or block-final distribution, so block packing drops no boundary likelihood.

Owned-fact routing. The route head scores every plan node against all true and hard-negative fact embeddings; $\mathcal{L}_{\text{route}}$ gives equal aggregate weight to positive owned node–fact pairs and negatives. Reference facts are not ownership targets for this head.

Outline progress. $\mathcal{L}_{\text{progress}}$ is cross entropy between each block-final hidden state and the assigned plan node whose target paragraph occupies most of that block.

Fact writing. For every lane, target block, and fact query, a classifier predicts OWNER, REFERENCE, or ABSENT; $\mathcal{L}_{\text{write}}$ uses inverse-frequency class weights $w_c = n/(3n_c)$, so the many absent labels cannot define a trivial optimum, and a fact written by the wrong physical lane is penalized even if it appears somewhere in the combined answer.

Dynamic-note alignment. $\mathcal{L}_{\text{note}}$ is symmetric InfoNCE at temperature 0.07 between each quantized note and its pre-quantized projection; other lane/block writes in the packed document supply negatives.

Presentation order. Given matched lane scores s and teacher order $\pi = (\pi_1, \pi_2, \pi_3)$, the Plackett–Luce loss is

$$\mathcal{L}_{\text{order}} = -\log \frac{e^{s_{\pi_1}}}{\sum_j e^{s_j}} - \log \frac{e^{s_{\pi_2}}}{e^{s_{\pi_2}} + e^{s_{\pi_3}}}. \quad (8)$$

Finite-channel regularization. For continuous subvector u and selected code e , $\mathcal{L}_{\text{commit}} = \|u - \text{sg}[e]\|_2^2$ and $\mathcal{L}_{\text{codebook}} = \|\text{sg}[u] - e\|_2^2$, and $\mathcal{L}_{\text{usage}}$ is the normalized deficit between maximum and empirical marginal code entropy in each product codebook, discouraging collapse without increasing channel capacity.

5.3 Combined objective

After plan matching, the canonical full objective is

$$\begin{aligned} \mathcal{L} = & 1.00\mathcal{L}_{\text{tok}} + 1.00\mathcal{L}_{\text{plan}} + 1.00\mathcal{L}_{\text{route}} + 0.50\mathcal{L}_{\text{progress}} \\ & + 1.00\mathcal{L}_{\text{write}} + 0.25\mathcal{L}_{\text{note}} + 0.10\mathcal{L}_{\text{order}} \\ & + 0.25\mathcal{L}_{\text{commit}} + 1.00\mathcal{L}_{\text{codebook}} + 0.10\mathcal{L}_{\text{usage}}. \end{aligned} \quad (9)$$

These coefficients are a registered starting configuration, not an asserted optimum; a pre-compiled one-factor architecture screen (Appendix A) varies capacity, channel, and optimizer factors before any selected interaction is tested in a separate registered run.

6 Curriculum and Optimization

Training separates capability questions that a single end-to-end run would confound; Table 2 gives the four stages. Every training example randomly maps its three semantic plans and target sections onto the three physical decoders, with all six permutations eligible across examples: the physical stacks may diverge numerically, but none can rely on a fixed semantic job, and an evaluation-time plan swap must move ownership with the plan while prompt and weights stay fixed.

Stage	Steps and plan	Trainable components	Scientific question
0	0–3,749; oracle outline	Three upper decoder banks, Plan-KV, SNC, semantic heads, note writer	Can the executor realize a correct three-way decomposition when the plan is given?
1	3,750–9,999; learned outline	Planner only; executor and semantic heads frozen	Can the planner reproduce an unordered teacher plan set through a fixed executor interface?
2	10,000–24,999; learned outline	Planner, upper decoder banks, Plan-KV, SNC, semantic heads, note writer	Can planning and execution adapt jointly over a frozen knowledge trunk?
3	25,000–49,999; learned outline	Same topology as Stage 2	Does continued joint training improve the same mechanism without a second path?

Table 2: Canonical training curriculum. Stage-specific loss weights disable objectives whose required module is frozen.

The shared lower trunk remains frozen in all stages. Physical branch weights keep FP32 optimizer masters with BF16 grouped computation; packed teacher forcing processes one 32-token block per lane in a single physical-frontier call, then publishes all active block notes. AdamW uses gradient clipping, cosine decay, and stage-aware module manifests. Checkpoints bind the trunk model and revision, fork layer, decoder count, coordination condition, optimizer, scheduler, step, and stage, so a checkpoint cannot silently load into a different architecture or a different bus/self-only condition.

7 Evaluation Protocol and Current Status

7.1 Questions, interventions, and gates

The evaluation is designed to prevent one success from standing in for another. Table 3 lists the principal questions; all confidence intervals are computed over documents rather than individual facts.

Question	Conditions	Preregistered evidence
Executor capability	Oracle teacher plan	Owner-fact recall lower bound $\geq 90\%$; unauthorized-leakage upper bound $\leq 10\%$; outline completion $\geq 90\%$; every lane long-form.
Planner capability	Oracle vs. learned vs. zero plan	Learned owner recall retains $\geq 80\%$ of oracle; zeroing all plan nodes drops owner recall by ≥ 30 percentage points.
Plan causality	Learned plan vs. physical D_1/D_2 plan swap	Swapped outputs retain $\geq 80\%$ of learned-plan owner recall at moved addresses; moved-address scoring beats deliberately unmoved scoring by ≥ 30 points.
Dynamic communication	Bus vs. dynamic-note removal	Document-bootstrap NLL change at exact dependency vs. nondependency tokens and their difference; Plan-KV intact in both conditions.
Sibling information	Bus vs. parameter-matched self-only memory	Owner/reference realization, delayed-reference timing, duplication, and prose quality under identical plan paths and training budgets.
Parallel execution	One lane; three serial lanes; packed frontier	Decode rounds, complete-answer latency, aggregate tokens/s, prefill and decode time, peak memory, per-round call and cache traces.

Table 3: Evaluation questions and preregistered evidence. Thresholds are gates for future measurements, not current results.

7.2 Fact ownership and long-form quality

Free generation stops each lane at its own EOS or at 1,000 new tokens. A valid lane contains at least 700 tokens and four paragraphs with no more than 10% of sentences shorter than eight words, rejecting the short-sentence failure mode directly. Fact presence is not exact string identity: a pinned NLI cross-encoder scores each atomic claim against sentence-level evidence in each physical lane, with a threshold calibrated once on disjoint teacher prose containing true and hard-negative claims, then frozen for held-out generation. The automatic score is a screen only. Every condition-lane-fact decision is exported to a blinded queue, judged by two independent annotators, and resolved by a distinct third adjudicator; final owner recall, reference recall, unauthorized leakage, and hard-negative rates use the adjudicated judgments.

7.3 Causal communication measurement

Teacher-forced communication evaluation reruns the exact target with dynamic notes removed and persistent Plan-KV unchanged. Retokenization has already marked every token overlapping a required receiver quote and verified that its source block is visible under the one-block delay. For token q , define

$$\Delta_q = \text{NLL}_{\text{no-dynamic}}(q) - \text{NLL}_{\text{bus}}(q). \quad (10)$$

The evaluator reports the document-level mean Δ on dependency tokens, on all other active tokens, and their selectivity difference: the mechanism predicts a larger benefit at annotated delayed dependencies, so a global likelihood change is insufficient. Destructive controls—mutating one producer’s code, exchanging note sources, scrambling note directions at matched norm, closing the note gate—diagnose address and payload sensitivity but do not replace the primary bus-versus-self-only comparison.

7.4 Scale control and status

The first model rung is a dense Qwen3-4B checkpoint [Qwen Team, 2025]. If its oracle-plan executor fails, data alignment, ownership labels, routing heads, optimization, cache behavior, and plan delivery are audited first; only if those pass is the same architecture repeated with a dense 14B trunk, asking whether the negative result is a trunk-capacity result rather than changing the scientific condition.

At the time of writing, the three physical decoder stacks, private caches, planner, Plan-KV, product-VQ notes bus, packed training rollout, counterfactual runtime, historical source filter, strict teacher schemas, and evaluators are implemented, and four manually audited long-form records exercise the data and trainer contracts, including sparse and zero-dependency cases. These establish executable plumbing only; no quality, coordination, or speedup result is reported in this manuscript.

8 Related Work

Prompted and learned semantic branching. Skeleton-of-Thought expands an elicited textual skeleton through parallel requests or batched decoding [Ning et al., 2024]; APAR post-trains fork control tokens and restricts attention over the paragraph tree [Liu et al., 2024]; PASTA learns a promise-and-sync annotation language so independent chunks decode asynchronously [Jin et al., 2025]. PDT shares the premise that reading order need not equal generation order, but makes three persistent upper decoder stacks part of the model, each conditioned on a continuous plan and a finite delayed latent channel.

Concurrent workers with high-bandwidth visibility. Hogwild! Inference lets concurrent model instances attend through a concurrently updated shared cache [Rodionov et al., 2025]; Group Think gives concurrent reasoning trajectories token-level mutual visibility [Hsu et al., 2025]. PDT studies the opposite communication regime: private causal caches, an explicit one-block barrier, and 32 learned payload bits per lane block, which makes causal attribution and channel capacity directly measurable.

Non-autoregressive and multi-token generation. Non-autoregressive translation predicts many target positions in parallel, trading away the left-to-right factorization and often relying on distillation [Gu et al., 2018, Kim and Rush, 2016]. Speculative decoding accelerates sampling while preserving the target distribution [Leviathan et al., 2023], Medusa adds multiple decoding heads [Cai et al., 2024], EAGLE speculates in feature space [Li et al., 2024], and multi-token prediction trains independent future-token heads above a shared trunk [Gloeckle et al., 2024]. PDT instead gives each head a complete, persistent autoregressive section that intentionally differs from the sequence a single left-to-right decoder would have sampled.

Set prediction, discrete latents, and external memory. Attention over learned seed queries yields set-valued outputs [Lee et al., 2019]; product vector quantization supplies a differentiable finite alphabet [van den Oord et al., 2017]; retrieval-augmented generation couples a parametric generator to external non-parametric memory [Lewis et al., 2020]. PDT’s plan and notes are internal, document-specific state derived from the current source and generated blocks.

9 Discussion and Limitations

Physical decoders without constant communication. The notes bus need not fire on every example for the physical split to matter. Three independent upper stacks can adapt to concurrent long-form trajectories, maintain private causal summaries, and follow different persistent plans without forcing all three roles through one upper parameterization; the model still emits three tokens per round when the optimal message is uninformative. Random semantic-to-physical assignment keeps this from collapsing into fixed experts: each stack may learn *trajectory capacity*, while its subject matter comes from the assigned plan. The self-only and no-dynamic controls price the physical factorization; the bus ablation prices sibling communication alone.

Persistent attention rather than a latent seed. A single additive vector can be ignored, overwritten, or entangled with lane state. Plan-KV instead exposes each outline node as addressable memory at every upper layer and token, asking the model to learn queries into a persistent structured memory rather than assuming instructions correspond to a linear direction in the residual stream. Complete-plan swaps then test causality: if assignment is real, ownership must move while physical weights stay fixed.

Offline and fixed-corpus deployment. Because the planner and notes bus operate entirely on local hidden states, the architecture is compatible with disconnected or fixed-corpus systems (onboard manuals, operating procedures, emergency protocols) where an online retrieval service [Lewis et al., 2020] is unavailable. This is a deployment hypothesis, not evidence of transfer: safety-critical use would require domain data, expert validation, failure-mode testing, and a separate safety case.

Falsifiability. The design is weakened by any of the following outcomes:

- the oracle-plan executor cannot achieve reliable lane ownership: three physical stacks do not by themselves solve assignment;
- the learned planner stays far below a sufficient oracle executor: decomposition is the bottleneck;
- plan swaps leave facts at their original physical addresses: branches learned fixed roles or ignore Plan-KV;
- dynamic-note removal has no selective effect on annotated dependency tokens: the bus is unused or unnecessary;
- the self-only control matches the bus on dependency-sensitive held-out data: sibling communication adds no measured value;
- grouped execution fails to reduce practical complete-answer latency: algorithmic span is not realized on the tested hardware.

These outcomes are separable: a failed notes bus does not erase the physical parallel frontiers, but it falsifies the stronger claim that a finite latent channel supports useful cross-lane coordination.

Limitations. Three independent upper stacks increase parameters, optimizer state, and upper-layer memory traffic; the canonical 24/12 split is a test point, not proof that half-depth branching is optimal; and Equation 6 must not be quoted as measured latency. The initial evidence domain is English historical exposition, chosen because facts, chronology, citations, and multi-section structure can be audited—code, mathematics, fiction, and tightly sequential reasoning may admit different decompositions or none at all. Wikipedia assessments and citations are filters, not guarantees; teacher-generated facts and plans can introduce errors under exact provenance schemas; and BGE targets, NLI screens, and product-VQ codes are model-dependent representations, which is why blinded human adjudication remains in the protocol. Finally, the one-block barrier excludes useful same-round interaction by design, and a 32-bit message may be too narrow, while a wider or more frequent channel may erase the constraint under study; the architecture screen measures this trade-off but cannot remove it.

10 Conclusion

The Parallel Decoder Transformer turns long-form parallel generation into a model architecture rather than an external scheduling trick. A frozen shared lower trunk supplies common language and world knowledge; a continuous planner produces an unordered set of three outlines; three independently parameterized upper stacks own private causal frontiers; persistent Plan-KV supplies structural assignment; and a finite delayed notes bus provides the only lawful path for cross-lane influence—three tokens per synchronization round without violating autoregression inside any section. Whether the architecture learns the intended behavior is the empirical question the registered protocol separates into executor capability, planner decomposition, plan causality, and dynamic communication; those experiments are in progress. The contribution of this paper is the precise, implementable, falsifiable design against which the eventual result can be judged.

A Canonical Instantiation and Registered Training Configuration

The first implementation uses `Qwen/Qwen3-4B-Instruct-2507` at a pinned revision [Qwen Team, 2025]: 36 decoder layers, hidden width 2,560, layers 0–23 frozen shared, layers 24–35 initializing each of the three physical upper stacks. Table 4 records the canonical dimensions. The physical stack keeps FP32 master parameters with BF16 grouped compute views; frozen lower-layer execution is outside autograd, and exact causal grouped-query attention is evaluated in bounded query tiles with activation recomputation.

Component	Canonical value
Physical decoder lanes	3
Shared / physical upper layers	24 / 3×12
Planner nodes per lane	8
Planner width / layers / heads	512 / 2 / 8
Plan and notes memory width	256
Plan/SNC attention width / heads	512 / 8
Target block size	32 tokens
Dynamic codebooks / entries	4 / 256
Logical dynamic capacity	32 bits per lane block
Dynamic history	16 blocks
Plan and notes outer gate initialization	−4.0
Shared frozen parameters	2,811,298,304
Physical upper-stack parameters	3,633,509,376
Planner and coordination parameters	84,132,905
Total materialized parameters	6,528,940,585

Table 4: Canonical architecture configuration. Parameter counts are an implementation census, not a performance result.

The registered optimizer is AdamW with learning rate 2×10^{-4} , weight decay 0.01, 1,250 warmup steps, cosine decay, gradient-norm clipping at 1.0, batch size one document, and gradient accumulation over 16 documents. The 16,384-token planner-prompt maximum is a fail-fast ceiling; source documents are never truncated to meet it. The architecture screen contains 25 one-factor variants at three independent seeds, varying model scale, upper-stack depth, planner depth and feed-forward width, notes and attention width, product-code shape, gate initialization, learning rate, weight decay, and bus versus parameter-matched self-only memory; it is compiled before training, and any interaction study is registered after the screen rather than selected post hoc.

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